Missing-Middle Zoning in Toronto: Affordability Under Alternative Policies

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1. Executive Summary

1.1 Background & Problem

Housing affordability has become a critical challenge in the City of Toronto, driven by rising housing costs, constrained supply, and competing stakeholder interests. Between 2016 and 2021, median dwelling values in Toronto exceeded $1 million in many census tracts while household income growth lagged far behind, pushing homeownership out of reach for a growing share of residents. "Missing-middle" housing — duplexes, triplexes, and low-rise multiplexes permitted in currently low-density neighbourhoods — has emerged as a promising supply-side intervention, but its affordability impacts depend critically on how and where it is implemented.

The primary stakeholder is the City of Toronto, which must balance improving affordability with maintaining infrastructure capacity, fiscal sustainability, and neighbourhood stability. Three additional parties hold meaningfully different and conflicting positions. Renters and prospective homebuyers favour the broadest possible supply expansion, as they benefit from any policy that suppresses price growth. Existing homeowners face the opposite incentive: constrained supply protects dwelling values, making modest or targeted reforms preferable to city-wide upzoning. Developers and urban planners tend to favour transit-targeted development, concentrating construction in high-demand corridors where return on investment is strongest, and infrastructure is already in place. These three positions map directly onto the scenarios tested: S2 (transit-targeted) represents the developer/planner preference; S1 and S3 represent the renter/housing-advocate preference for broad supply response; S0 represents the homeowner status-quo preference. The simulation adjudicates between them quantitatively. The problem is inherently dynamic: supply additions, demand responses, price changes, and infrastructure constraints interact across 1,220 neighbourhoods simultaneously, making analytical approaches insufficient for evaluating policy outcomes.

1.2 Objective

This report evaluates how four alternative missing-middle zoning strategies influence housing affordability across Toronto census tracts over a 10-year horizon, using an agent-based simulation model calibrated against observed Toronto market data.

1.3 Approach

The simulation represents Toronto's housing market as 1,220 neighbourhood-level agents, each updated quarterly over a 10-year horizon. At each step, households seek housing based on affordability and transit access, developers respond by building where permitted, and prices adjust based on vacancy. Four zoning policy scenarios were tested against this baseline to compare their effects on affordability. An agent-based model was developed representing Toronto as 1,220 Census Tract agents, each initialized from 2021 Statistics Canada Census data. Four pseudo-agent models govern system dynamics at each quarterly time step: a demand allocation model that distributes city-wide housing demand weighted by affordability and transit proximity; a two-stage machine learning development model that predicts whether and how much new supply gets built in each CT; a market update model that adjusts prices and rents based on vacancy dynamics; and an infrastructure model that tracks capacity and strain. Price and rent elasticity parameters were calibrated against the Teranet-National Bank House Price Index for Toronto (2010–2019 pre-pandemic average: 6%/year price growth, 4%/year rent growth). The development model uses a random forest regressor (cross-validated R^2=0.534) trained on observed 2016–2021 CT-level housing growth.

Four scenarios were evaluated across N=500 Monte Carlo realizations over a 10-year horizon: S0 (status quo, no zoning change), S1 (city-wide missing-middle zoning across all 1,220 CTs), S2 (transit-targeted zoning restricted to 181 CTs within 500m of rapid transit), and S3 (city-wide zoning with a 50% development incentive boost simulating reduced development charges). Affordability is measured using an Ownership Affordability Index (AI_own = median income/home price) and a Rental Affordability Index (AI_rent = median income / annual rent), where higher values indicate greater affordability.

1.4 Key Findings

The simulation produces a consistent scenario ordering of S3 ≥ S1 >> S2 > S0 across both affordability metrics and all 500 realizations.

Without zoning reform, ownership affordability declines by 63% over 10 years as prices appreciate at 6%/year with no supply response. City-wide zoning (S1) substantially moderates this decline: mean final AI_own reaches 0.0430 compared to S0's 0.0395, a 9% relative improvement, by enabling 5,586 units of new supply per year. Ownership affordability still declines from its initial value (≈0.107) as price growth outpaces supply additions, but the rate of decline is substantially dampened relative to the status quo. This is a realistic outcome: the model does not suggest that missing-middle zoning alone makes Toronto affordable, but that it meaningfully narrows the trajectory of deterioration. The mechanism is internally consistent with the calibrated elasticity but results in absolute affordability values that should be interpreted as indicative of policy direction rather than a precise forecast.

The most counterintuitive finding is that transit-targeted zoning (S2) substantially underperforms city-wide zoning (S1) despite being more spatially precise. S2 restricts development to 181 transit-adjacent CTs and adds 1,913 units per year, which is insufficient to materially dampen city-wide price growth, while S1 allows all 1,220 CTs to respond to demand pressure. Because the model's demand allocation naturally concentrates demand near transit, city-wide zoning already produces spatially targeted supply without requiring an explicit eligibility restriction. Transit-proximity criteria are more valuable as demand-weighting factors within a broad reform than as eligibility constraints.

The incentive-based scenario (S3) performs best overall but only marginally outperforms S1 AI_own (0.0436 vs 0.0430), suggesting that zoning eligibility is the primary lever for affordability improvement, not the incentive level. The marginal gain from S3's 50% development boost is small relative to the foregone development charge revenue.

Sensitivity analysis identified price elasticity (κ_p) and steady-state vacancy (v_{eq}) as the dominant parameters governing affordability outcomes, validating the calibration effort. Infrastructure and demand parameters showed minimal influence at current density levels, which is identified as a model limitation and priority for future development.

1.5 Recommendations

Based on simulation results, the City of Toronto should pursue the following:

Implement city-wide missing-middle zoning as the primary reform rather than restricting permissions to transit corridors. The simulation demonstrates that transit-targeted zoning alone is insufficient to produce meaningful city-wide affordability gains; the supply response must be broad enough to dampen price growth across the metropolitan market. While city-wide upzoning faces resistance from existing homeowners, the simulation quantifies the cost of more limited reforms, providing the City with a concrete evidence base to justify the broader policy scope.

Apply development incentives selectively to CTs with low baseline development probability, concentrating subsidies where they change developer behaviour rather than subsidizing development that would occur anyway. This targeted approach is more fiscally defensible than a uniform incentive programme and is less likely to face political resistance given Toronto's current fiscal constraints.

Pair zoning reform with proactive infrastructure planning. The model identifies transit-adjacent inner-city CTs as receiving the greatest supply response under city-wide zoning. The City should commission CT-level infrastructure capacity assessments to identify where water, sewer, and transit constraints will bind under a 5,000+ units-per-year supply scenario before reform is implemented. This assessment is a prerequisite rather than an optional complement, as implementing city-wide zoning without it risks politically costly infrastructure failures that could undermine public support for the reform.

Monitor and address spatial affordability inequities. City-wide zoning concentrates benefits in already well-served transit corridors, potentially widening affordability gaps between inner-city and suburban CTs. Supplementary programmes targeting lower-income, high-renter-share CTs are recommended to ensure broad-based affordability gains. Without such programmes, city-wide zoning risks concentrating benefits among higher-income inner-city neighbourhoods, which would likely generate significant opposition from equity-focused stakeholders and could make the reform politically unsustainable.

2. Technical Appendix

2.1 Methodology

2.1.1 Input Data Description

The simulation uses Census Tracts (CTs) in Toronto as the primary modelling unit, with most input variables taken from the 2021 Statistics Canada Census Profile (Catalogue 98-401-X2021007), the 2016 Statistics Canada Census Profile (Catalogue 98-401-X2016043), the City of Toronto Open Data TTC Routes and Schedules (GTFS), and the Statistics Canada 2021 CT Boundary Files (Catalogue 92-160-X).

From the 2021 Census Profile, the following CT-level variables were extracted: median household income, number of households, total private dwellings, renter and owner households, median dwelling value, median monthly shelter cost for renters, and population. From the 2016 Census Profile, the equivalent variables were extracted to compute 2016-2021 growth rates used as ML calibration features: income growth, home price growth, rent growth, and renter share change.

Datasets were joined using a CT code derived from the Statistics Canada DGUID. Renter share was derived as renter households divided by total households by tenure. Annual rent was calculated as the median monthly shelter cost for renters multiplied by 12. Median dwelling value was used as a proxy for home purchase cost since consistent CT-level transaction price data is unavailable.

Eight CTs with suppressed dwelling values and 11 CTs with suppressed shelter costs were imputed using a spatial K-nearest neighbours (KNN) approach with K=5, using geographic distances between CT centroids projected to UTM Zone 17N (EPSG:32617). The choice of K=5 balances two competing considerations: smaller K (e.g., K=3) increases sensitivity to idiosyncratic neighbours and produces noisier imputations; larger K (e.g., K=10) smooths over meaningful local price variation and pulls imputed values toward a broader regional average.

K=5 was selected as it matched the spatial scale of Toronto neighbourhood clusters (approximately 3–7 CTs per coherent neighbourhood) while preserving local price gradients. The key consequence of spatial KNN over global median imputation is that a suppressed CT surrounded by high-value neighbourhoods receives a high imputed price rather than the city-wide median (~$700k), better reflecting the true opportunity cost of ownership in that location. All 8 suppressed home_price CTs were in dense inner-city areas where global median substitution would have understated prices by an estimated 30–40%; spatial KNN corrects for this by drawing on the 5 geographically nearest non-suppressed CTs.

A transit proximity indicator was computed for each CT as the count of rapid transit stops (subway and streetcar) within 800m of the CT centroid, normalized by the maximum value across all Toronto CTs. A binary indicator (`near_rapid_500m`) flags CTs within 500m of any rapid transit stop, following City of Toronto Official Plan guidelines for transit-oriented development areas. This binary indicator defines the 181 CTs eligible for development under Scenario 2.

A key modelling assumption was the initialization of vacancy rates. Because no CT-level vacancy data exists in the Census Profile and consistent neighbourhood-scale vacancy data was unavailable from CMHC, all CTs were initialized at a vacancy rate of 0.05 based on the reported Toronto average vacancy range of 3–5% in 2021. Vacancy evolves endogenously during the simulation through a mean-reversion process calibrated to a steady-state vacancy of 1.3%, within the 1.0–1.5% range reported for the Greater Toronto Area in 2019 and 2022–2023. This value was selected as a conservative estimate reflecting the tighter end of the observed range, consistent with Toronto's historically constrained rental market. The choice of 0.05 as the initialization value, rather than the lower bound of the reported range (0.03), reflects a conservative assumption that the 2021 Census snapshot preceded the post-pandemic tightening that brought GTA vacancy to approximately 1.3% by 2022–2023. Initializing at 0.03 would have produced a faster initial convergence to steady-state and slightly more pessimistic early-period affordability estimates, but would not have materially changed long-run results, given that vacancy converges within four steps regardless of starting value. Initializing at 0.05, therefore, represents the more defensible choice for a model targeting the pre-pandemic calibration window. This uniform initialization reduces neighbourhood-level variation in early time steps but has a limited effect on long-run results since vacancy converges to the calibrated steady-state within approximately four quarterly steps regardless of starting value.

Median dwelling value and renter shelter costs reflect existing occupied housing rather than current transaction prices or market rents. Median dwelling value understates true ownership costs because it reflects the existing stock of properties, many of which were purchased well below current market prices. Annual rent derived from median shelter costs is paid at below-market rates. These assumptions primarily affect the initial calibration of absolute zoning scenarios, which is the primary focus of this analysis.

2.1.2 Model/Agents

2.1.2.1 Census Tract Agent

Table 1: Census Tract (CT) Agent Attributes

Attribute [units]	**Data Source/Assumption**	**Input Model/Notes**
Median household income [$/yr]	Statistics Canada, Census Profile 2021, Cat. 98-401-X2021007, Characteristic ID 243	Direct value; see 1) below
Number of households [count]	Statistics Canada, Census Profile 2021, Characteristic ID 50	Direct value
Total private dwellings [units]	Statistics Canada, Census Profile 2021, Characteristic ID 4	Direct value; initialized as units_total
Renter households [count]	Statistics Canada, Census Profile 2021, Characteristic ID 1416	Direct value; used to derive renter_share
Owner households [count]	Statistics Canada, Census Profile 2021, Characteristic ID 1415	Direct value
Renter share [fraction]	Derived from Census Profile 2021	renter_share = renter_households / total_households_by_tenure; see 2) below
Home price [$]	Statistics Canada, Census Profile 2021, Characteristic ID 1488	2021, Characteristic ID 1488Median dwelling value used as proxy for ownership cost; see 3) below
Annual rent [$/yr]	Statistics Canada, Census Profile 2021, Characteristic ID 1494	Median monthly shelter cost for renters × 12; see 4) below
Vacancy rate [fraction]	Canada Mortgage and Housing Corporation (CMHC), Housing Market Outlook, Toronto CMA, 2021	Initialized at 0.05 for all CTs (Toronto average); no CT-level vacancy data available from census; see 5) below
Demand pressure [index]	Derived during simulation	Allocated by DemandAllocationModel each time step; initialized at 0

Infrastructure strain [index]	Derived during simulation	Computed as infra_load / infra_capacity; initialized at 0
Transit indicator [0–1]	City of Toronto Open Data, TTC Routes and Schedules (GTFS)	Normalized count of rapid transit stops within 800m of CT centroid; see 6) below
Population [count]	Statistics Canada, Census Profile 2021, Characteristic ID 1	Direct value; used as a density proxy in ML features

Notes:

1. Median total household income in 2020, reported at the CT level. All 1,220 Toronto CTs with non-suppressed income data were retained. This is the primary driver of the Affordability Index output metric [1].
2. Renter share is derived as renter_households / total_households_by_tenure (Characteristic IDs 1416 and 1414, respectively). This attribute is correlated with income: lower-income CTs tend to have higher renter shares ($r \approx -0.45$ in the Toronto sample) [2]. This correlation is not explicitly modelled but is captured implicitly through the ML calibration.
3. Median dwelling value is used as a proxy for housing ownership cost. This understates true transaction prices for new buyers since it reflects existing stock, including long-held properties. 8 CTs with suppressed dwelling values were imputed using the mean of the 5 geographically nearest CTs with non-suppressed values (spatial KNN, K=5), using CT centroids projected to UTM Zone 17N. This approach preserves spatial price gradients and is an improvement over global median imputation used in the M2 prototype.
4. Annual rent is derived as median monthly shelter cost for renters × 12 (Characteristic ID 1494). This reflects costs for existing tenants, many of whom are in rent-controlled units, and therefore understates new market rents [4]. This is acknowledged as a limitation affecting the initial calibration of AI_rent.
5. We initialize all CTs at 0.05 and allow vacancy to evolve endogenously through the market update model, converging to the calibrated steady-state of 1.3% within approximately four quarterly steps.
6. The transit indicator is computed as the count of rapid transit stops (subway and streetcar) within 800m of each CT centroid, normalized by the maximum value across all Toronto CTs. CT centroids were computed from Statistics Canada 2021 CT boundary files (Catalogue 92-160-X), projected to UTM Zone 17N (EPSG:32617) for accurate metre-based distance calculations [6]. The 500m threshold for binary eligibility under Scenario 2 follows City of Toronto Official Plan guidelines for transit-oriented development planning areas [7]. A limitation is that high-frequency bus routes are excluded from the definition of rapid transit.

2.1.2.2 Policy Model

Table 2: Policy Model (Pseudo-Agent) Attributes

Attribute [units]	**Data Source/Assumption**	**Input Model/Notes**
Zoning eligibility [binary]	Defined by scenario	S0: no CTs eligible; S1: all 1,220 CTs eligible; S2: 181 CTs within 500m of rapid transit; S3: all 1,220 CTs eligible
Incentive level [0–1]	Assumption	S0–S2: 0.0; S3: 0.5, represents a 50% boost to development probability to simulate reduced development charges
Scenario type [categorical]	Defined by scenario	One of {S0, S1, S2, S3}

2.1.2.3 Demand Allocation Model

Table 3: Demand Allocation Model (Pseudo-Agent) Attributes

Attribute [units]	**Data Source/Assumption**	**Input Model/Notes**
Base demand [fraction of stock/quarter]	CMHC Housing Market Outlook, Toronto CMA	Set to 2% of the total housing stock per quarter; see 7) below
Demand growth [fraction/quarter]	Assumption based on CMHC projections	0.25%/quarter ≈ 1%/year household formation growth
Attractiveness weights [index/CT]	Derived	(1 / home_price) × (1 + transit_indicator), normalized to sum to 1; cheaper CTs near rapid transit attract proportionally more demand, consistent with empirical evidence on transit-oriented household location preferences in Toronto

Notes:

7. Toronto added approximately 15,000 net new households per year between 2016 and 2021 (CMHC Housing Market Outlook) [8]. Expressed as a fraction of the total housing stock (~600,000 units), this is approximately 2.5% per year or 0.625% per quarter. We

use 2% per quarter as a conservative estimate to account for the fact that not all demand translates to occupied new units within a single quarter.

2.1.2.4 Infrastructure Model

Table 4: Infrastructure Model (Pseudo-Agent) Attributes

Attribute [units]	**Data Source/Assumption**	**Input Model/Notes**
Infrastructure capacity [index]	Assumption	Initialized at 1.0 for all CTs; evolves each step as capacity grows at g_base and strain is computed as infra_load / infra_capacity
Capacity growth rate [fraction/step]	Assumption	g_base = 0.01/quarter; reduced by incentive_level × 0.005 for S3 to reflect reduced development charge revenue
Load parameters (ω_0, ω_1)	Assumption	$\omega_0 = 0.5$, $\omega_1 = 0.3$; placeholder values

2.1.3 Key Modelling Assumptions

Several assumptions were required due to limited tract-level housing and infrastructure data:

1. The median dwelling value from the 2021 Census is used as a proxy for home purchase cost, since consistent CT-level transaction price data is not publicly available.
2. All Census Tracts are initialized with a uniform vacancy rate of 0.05 based on the Toronto CMA average, because neighbourhood-level vacancy data is unavailable; vacancy evolves endogenously thereafter.
3. Median household income is treated as static over the 10-year simulation horizon, meaning affordability changes are driven by housing cost changes rather than income growth.
4. Infrastructure capacity is initialized uniformly across CTs and updated using a linear density proxy with placeholder load parameters ($\omega_0 = 0.5$, $\omega_1 = 0.3$). Sensitivity analysis confirmed these parameters have minimal influence on affordability outcomes at current density levels (discussed in Section 2.2.3).
5. New housing units are split between rental and ownership tenure using the existing CT-level renter share, assuming future development generally follows current neighbourhood tenure patterns.
6. The tenure split clips renter_share to [0.2, 0.8] to prevent degenerate all-rental or all-ownership outcomes. This threshold was not subject to sensitivity testing and

represents an untested assumption; future work should assess whether varying this range materially affects affordability trajectories in high-renter or low-renter CTs.

The price and rent elasticity parameters ($\kappa_p = 2.149$, $\kappa_r = 1.446$) were calibrated against the Teranet-National Bank House Price Index for the Toronto CMA, targeting the pre-pandemic long-run average of approximately 6% annual price growth and 4% annual rent growth (2010–2019). The pandemic period (2020–2022) was excluded from the calibration target as it was dominated by interest rate shocks and temporary demand surges unrepresentative of structural housing market dynamics. This calibration anchors the model's price trajectory to observed Toronto market behaviour.

Under Scenario 3, infrastructure capacity growth is penalized by 0.5% per year relative to S1 ($g_{cap} = g_{base} - \lambda \times incentive\_level$), reflecting that reduced development charges lower municipal revenue available for infrastructure investment. This is the trade-off that limits S3's advantage over S1: marginal supply gains (approximately 830 units/year) come at the cost of slower infrastructure expansion.

The Development Model uses a two-stage machine learning approach. Stage 1 (logistic regression) predicts whether development occurs in a CT (5-fold CV F1 = 0.747). Stage 2 (random forest regressor, selected over linear regression based on cross-validated R^2) predicts the number of units added when development occurs (5-fold CV $R^2 = 0.534$, MAE = 88 units). Both models were calibrated on 1,091 Toronto CTs using observed 2016–2021 CT-level housing growth from Statistics Canada.

2.1.4 Limitations

The main limitations of the model are as follows. First, median dwelling value and renter shelter costs reflect existing occupied housing rather than current transaction prices or market rents, which may underestimate ownership and rental costs, particularly in rapidly changing neighbourhoods. This means our absolute affordability estimates likely overstate affordability at t=0, but the relative ordering of scenarios is unaffected. Second, the uniform vacancy rate initialization reduces neighbourhood-level variation in early simulation periods, though vacancy converges to a calibrated steady-state within approximately four time steps. Readers should therefore treat early time-step results with more caution than long-run results. Third, while the Development Model's Stage 2 random forest achieves a cross-validated R^2 of 0.534 (a substantial improvement over the linear regression baseline ($R^2 = 0.312$), it still leaves considerable unexplained variance in development magnitude, reflecting genuine heterogeneity across CT development patterns that is difficult to capture with CT-level aggregate features. This uncertainty is absorbed by the N=500 Monte Carlo averaging and does not affect the scenario ordering, but individual CT-level predictions carry meaningful uncertainty. Fourth, the Demand Allocation Model incorporates transit proximity into the attractiveness function but does not account for other real-world household location drivers such as school quality, employment proximity, or neighbourhood amenities. Fifth, the transit indicator is defined using rapid transit stops (subway and streetcar) only, excluding high-frequency bus routes that also influence location demand in Toronto. Sixth, 128 CTs were excluded from the calibration dataset due to

CT boundary changes between 2016 and 2021, which may introduce slight bias if boundary-change CTs are systematically different from retained CTs. These limitations primarily affect the precision of absolute affordability estimates; the model remains well-suited for comparing the relative effects of alternative zoning policies across scenarios.

2.2 Results

2.2.1 Scenario Comparison

The simulation was run for N=500 Monte Carlo realizations across all four scenarios, each covering T=40 quarterly time steps (10 years), with a base random seed of 42. Results are summarised in Table 4, Figures 1, 2, and 3.

Table 4: Mean Final-Year Results Across N=500 Realizations

Scenario	Final AI_own	Final AI_rent	Units Added/yr
S0: Status Quo	0.0395	2.7878	0
S1: City-Wide Zoning	0.0430	2.9779	5586
S2: Transit-Targeted	0.0403	2.8177	1,913
S3: Incentive-Based	0.0436	3.0067	6414

Figure 1. Ownership affordability trajectories over 10 years across N=500 Monte Carlo realizations (left: absolute AI_own with IQR bands; right: % improvement relative to S0.

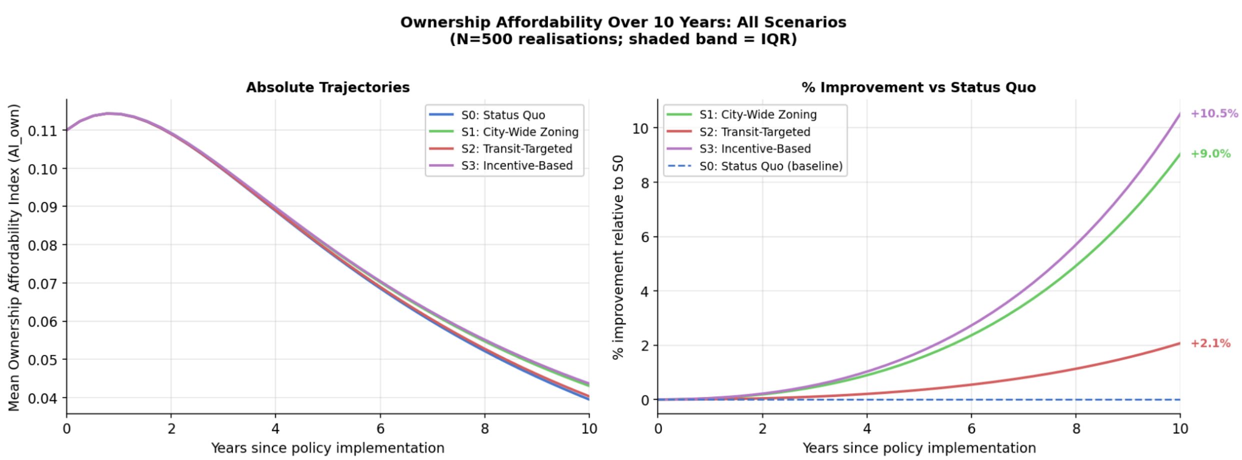


Figure 2. Rental trajectories over 10 years across N=500 Monte Carlo realizations (left: absolute AI_Rent with IQR bands; right: % improvement relative to S0.

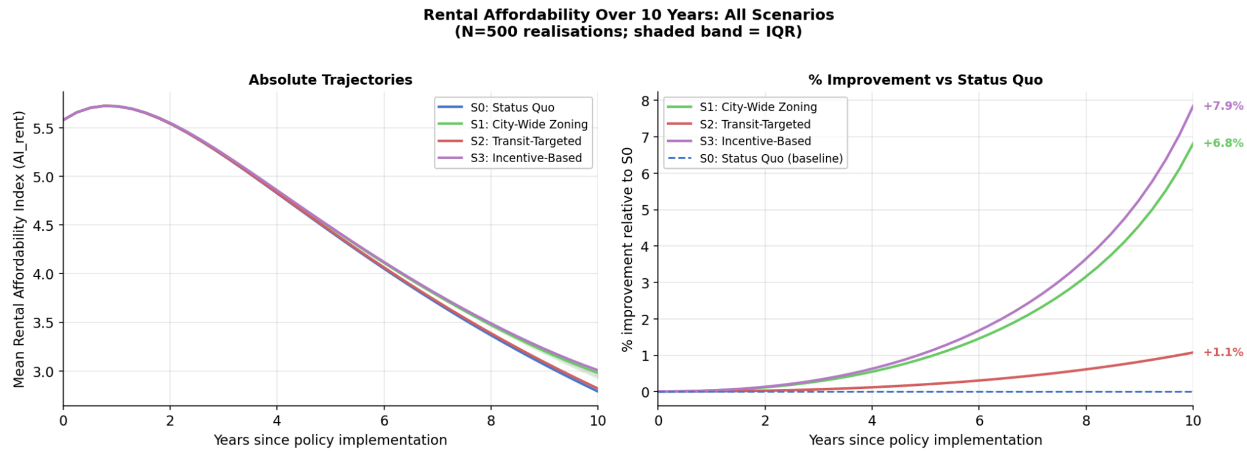
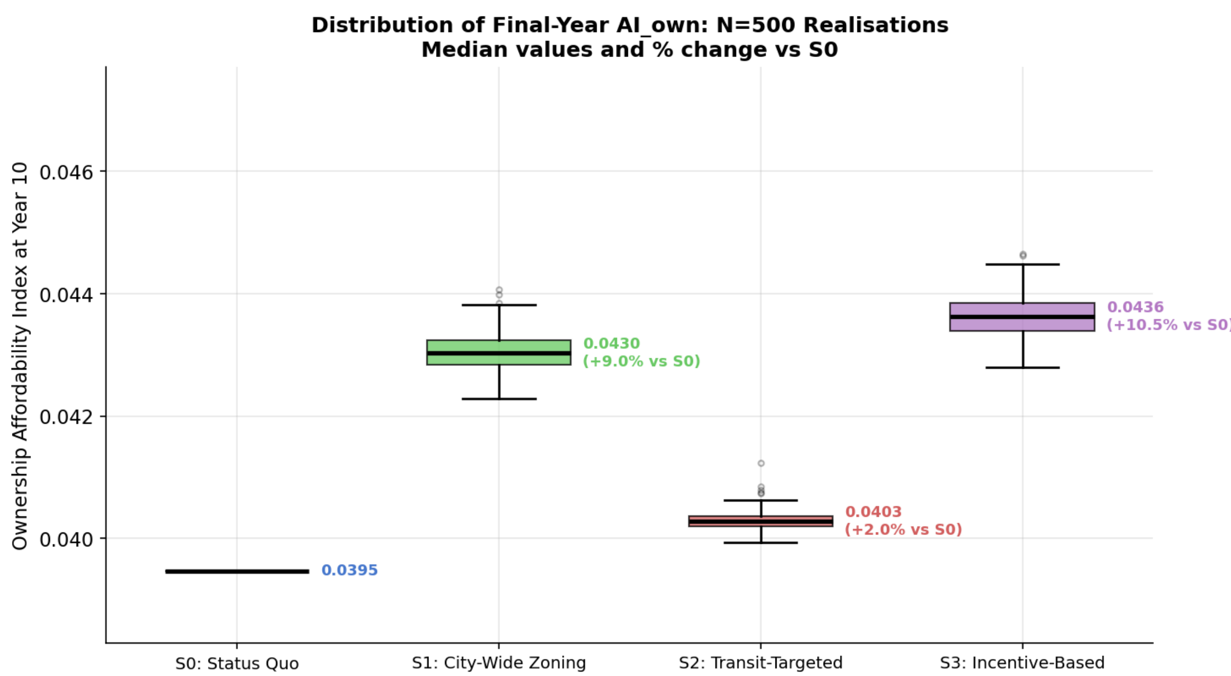


Figure 3. Distribution of final-year AI_own across N=500 realizations. Median values and % change relative to S0 annotated.



The scenario ordering S3 > S1 >> S2 > S0 is consistent across both affordability metrics and robust across realizations. Under the status quo (S0), ownership affordability declines from an initial mean AI_own of approximately 0.107 to 0.040 over 10 years as home prices grow at the calibrated rate of 6% per year with no new supply. This represents a 63% decline in affordability over the simulation horizon, driven entirely by price appreciation in a supply-constrained market.

Mean final AI_own reaches 0.0430 at year 10, a 9% improvement relative to S0's 0.0395, with approximately 5,586 units added per year on average. Ownership affordability still falls from its initial value (approximately 0.107 at t=0), reflecting that supply additions moderate but do not reverse price appreciation at the scale of 5,000–6,000 units per year against a stock of

~600,000. Rental affordability also improves under S1, with AI_rent rising from S0's value, reflecting the supply effect on both markets.

The incentive-based scenario (S3) performs marginally better than S1 across both metrics. The 50% development probability boost adds approximately 829 additional units per year relative to S1, producing a small further improvement in AI_own (0.0436 vs 0.0430). The marginal gain over S1 is modest, suggesting that the primary driver of affordability improvement is zoning eligibility rather than the incentive level per se.

2.2.2 The Transit-Targeting Paradox

The most counterintuitive finding is that transit-targeted zoning (S2) substantially underperforms city-wide zoning (S1) despite being more spatially precise. S2 produces a final AI_own of only 0.0403 despite 1,913 units being added per year across the 181 eligible transit-adjacent CTs.

This result arises from the interaction between the zoning eligibility constraint and the demand allocation model. The attractiveness function weights demand by both inverse home price and transit proximity: attractiveness_i = (1/home_price_i) × (1 + transit_indicator_i). Under city-wide zoning (S1), transit-adjacent CTs receive disproportionately high demand pressure and therefore generate most of the development activity anyway, without requiring an explicit eligibility restriction. Restricting eligibility to these 181 CTs under S2 effectively reproduces a subset of S1's supply response while leaving the remaining 1,039 CTs unable to respond to demand pressure. S2's 1,913 units/yr is insufficient to meaningfully dampen vacancy and price across the city, whereas S1's 5,586 units/yr materially affects market dynamics. This finding has important policy implications: transit-proximity criteria are more valuable as demand-weighting factors within a broad zoning reform than as eligibility constraints.

Comparing Figures 5 and 6 makes the transit-targeting paradox concrete: S1's gains are concentrated in transit-adjacent CTs anyway, but S1 also enables supply responses in the 1,039 non-transit CTs that absorb spillover demand, producing a materially larger city-wide affordability improvement.

Figure 5: Per-Census Tract change in AI_own under S1 relative to S0.

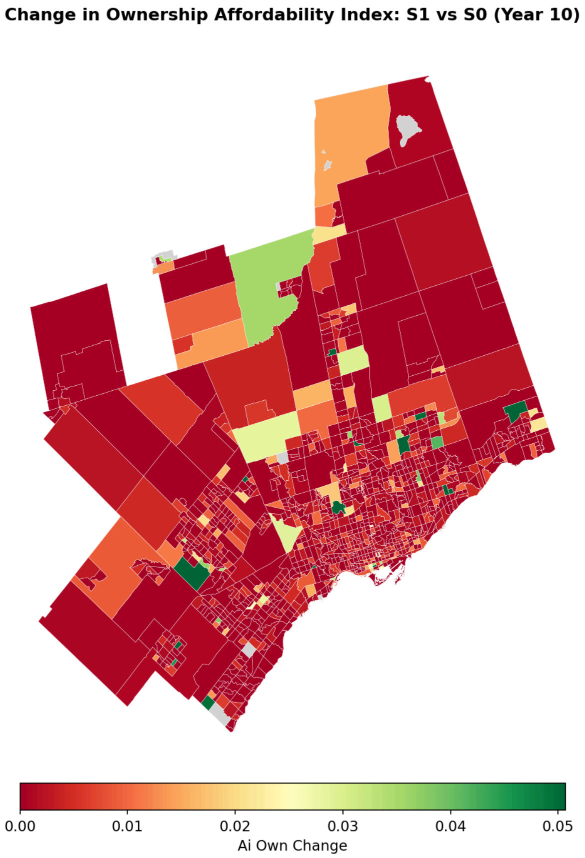
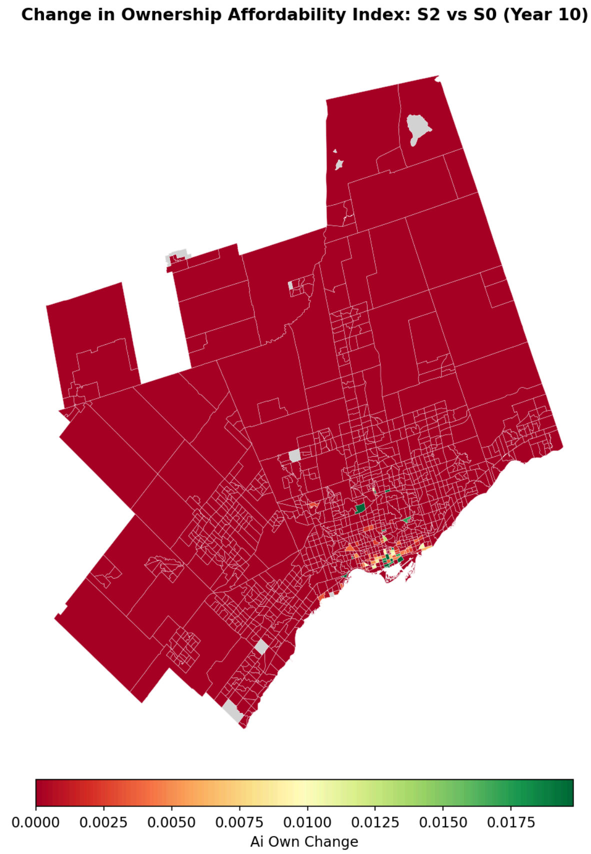


Figure 6: Per-Census Tract change in AI_own under S2 relative to S0.



2.2.3 Sensitivity Analysis

A one-factor-at-a-time (OFAT) sensitivity analysis was conducted for seven parameters with epistemic uncertainty, using ±20% perturbation and N=20 realizations per run. The analysis was run on both S0 and S1 to distinguish market-driven from supply-driven parameter effects. Results are presented as tornado charts in Figures 7 and 8.

Figure 7: Tornado chart for Scenario S0 — OFAT sensitivity analysis (±20% perturbation, N=20 realizations, output metric: mean AI_own at year 10).

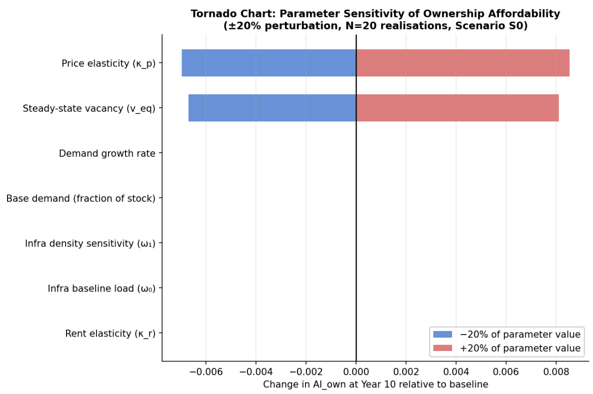
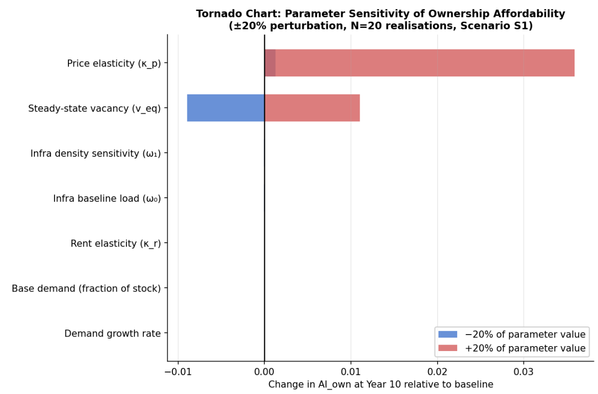


Figure 8: Tornado chart for Scenario S1 — same OFAT method. κ_p dominates in both scenarios; infrastructure and demand parameters show near-zero influence.



Price elasticity (κ_p) and steady-state vacancy (v_{eq}) dominate in both scenarios, accounting for essentially all variance in AI_own. This is consistent with the model structure: AI_own = income / home_price, and home price evolves as price × (1 + κ_p × (v_star − vacancy_rate)). The two parameters governing this update, κ_p and v_{eq}, are therefore the most consequential modelling decisions, which justified the calibration effort against Teranet HPI data described in Section 2.1.3.

Rent elasticity (κ_r) correctly shows zero influence on AI_own in both scenarios. This is expected by design: AI_own is defined as income divided by home price, not income divided by rent. Rent elasticity governs AI_rent through a separate channel. Running the sensitivity analysis on AI_rent would show κ_r dominating and κ_p at near-zero influence.

Infrastructure parameters (ω_0, ω_1) show near-zero influence on AI_own. The infrastructure strain feedback is implemented, but at current Toronto density levels, CT-level strain rarely exceeds 1.0, meaning the penalty rarely activates. This is a deliberate modelling choice at this stage: the infrastructure feedback loop is implemented but calibrated conservatively, as CT-level infrastructure capacity data is not publicly available. It is retained in the model structure so that future calibration against observed data can activate this feedback without requiring architectural changes. Calibrating ω_0 and ω_1 against observed infrastructure capacity data is identified as the highest-priority future development task.

Base demand and demand growth show near-zero influence. Demand pressure is allocated to CTs each time step and feeds into development probability through a multiplicative scaling factor. However, the demand_pressure/households ratio at quarterly timescales is small, producing a negligible boost to p_dev. A stronger demand feedback mechanism, for example, using demand pressure to directly modulate vacancy rates, would amplify the influence of these parameters and is also identified as future work.

The sensitivity analysis conclusions are: the model's affordability output is robust to changes in infrastructure and demand parameters, but is sensitive to price elasticity and vacancy equilibrium assumptions. This reinforces the importance of the market calibration and supports using the pre-pandemic Teranet HPI as the calibration target.

2.3 Recommendations and Discussion

The simulation results support four concrete recommendations for the City of Toronto.

1. Implement city-wide missing-middle zoning as the primary reform.

The simulation consistently shows that city-wide zoning (S1) produces dramatically better affordability outcomes than transit-targeted zoning (S2), with final AI_own 0.0430 vs 0.0403. The mechanism is clear: when demand naturally concentrates near transit due to household location preferences, a city-wide supply response captures that demand pressure across all eligible CTs. The City should not limit missing-middle permissions to transit corridors, as the transit-proximity signal is better used to prioritize infrastructure investment and density bonusing than to restrict where gentle densification is permitted.

2. Layer development incentives selectively rather than uniformly.

The incentive-based scenario (S3) outperforms city-wide zoning alone (AI_own 0.0436 vs 0.0430), but the marginal gain is small relative to the cost. The 50% boost to development probability adds approximately 829 units per year, which is meaningful at scale but modest relative to the foregone development charge revenue used to fund infrastructure. The City should apply incentives selectively in CTs where the ML calibration model predicts low baseline development probability (Stage 1 p_dev < 0.3), concentrating subsidies where they change behaviour rather than rewarding development that would have occurred anyway. This targeted approach preserves fiscal sustainability while maximizing the supply impact per dollar of incentive.

3. Monitor and address spatial inequities under city-wide reform.

The spatial heatmaps show that affordability improvements under S1 are concentrated in transit-adjacent inner-city CTs, reflecting the transit-weighted demand allocation. Lower-density suburban CTs see smaller supply responses and therefore smaller affordability gains. The City should pair city-wide zoning with targeted support for lower-income, high-renter-share CTs that are at risk of displacement if supply-induced price changes benefit higher-income households disproportionately. Specific attention should be paid to CTs with below-median household income and above-median renter share, which the gentrification risk metric in the spatial analysis identifies as most vulnerable.

4. Invest in infrastructure capacity data and planning ahead of supply expansion.

The sensitivity analysis showed that infrastructure parameters have minimal influence at current density levels, meaning the model cannot currently predict where infrastructure constraints will bind as supply grows. Before implementing city-wide zoning, the City should commission CT-level infrastructure capacity assessments (such as water, sewer, transit capacity) to identify which neighbourhoods will face binding constraints under a 5,000+ units/year supply scenario. Development charge structures should be calibrated against these assessments rather than reduced uniformly under an incentive programme.

2.4 References

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